

1. (Question 1). Let X be a discrete random variable with the given probability mass function with $0 \leq \theta \leq 1$ as a parameter:

x	0	1	2	3
$P(X = x \theta)$	$2\theta/3$	$\theta/3$	$2(1 - \theta)/3$	$(1 - \theta)/3$

We have the following i.i.d. sample $\mathbf{x} = (3, 0, 2, 1, 3, 2, 1, 0, 2, 1)$ from X and we want the maximum likelihood estimate of θ . Let x^t denote the t -th observation, for $t = 1, \dots, N$, where here $N = 10$, so that $\mathbf{x} = (x^1, \dots, x^N)$. Since the observations are independent, the likelihood of the parameter θ is

$$\begin{aligned}
 l(\theta | \mathbf{x}) &= \prod_{t=1}^N p(x^t | \theta) \\
 &= p(3 | \theta) \cdot p(0 | \theta) \cdot p(2 | \theta) \cdot p(1 | \theta) \cdot p(3 | \theta) \\
 &\quad \cdot p(2 | \theta) \cdot p(1 | \theta) \cdot p(0 | \theta) \cdot p(2 | \theta) \cdot p(1 | \theta) \\
 &= \frac{1 - \theta}{3} \cdot \frac{2\theta}{3} \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \cdot \frac{1 - \theta}{3} \\
 &\quad \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \cdot \frac{2\theta}{3} \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \\
 &= \left(\frac{2\theta}{3}\right)^2 \left(\frac{\theta}{3}\right)^3 \left(\frac{2(1 - \theta)}{3}\right)^3 \left(\frac{1 - \theta}{3}\right)^2 \\
 &= \frac{2^5}{3^{10}} \theta^5 (1 - \theta)^5.
 \end{aligned}$$

We can optimize via the log likelihood and our desired MLE for θ is $\hat{\theta} = \arg \max_{\theta} (L(\theta | \mathbf{x}))$ so consider

$$\begin{aligned}
 L(\theta | \mathbf{x}) &= \log(l(\theta | \mathbf{x})) \\
 &= \log\left(\frac{2^5}{3^{10}} \theta^5 (1 - \theta)^5\right).
 \end{aligned}$$

Applying rules of logarithms we have

$$L(\theta | \mathbf{x}) = \log\left(\frac{2^5}{3^{10}}\right) + 5 \log \theta + 5 \log(1 - \theta).$$

Taking the derivative of L with respect to θ

$$\begin{aligned}\frac{dL}{d\theta} &= 0 + 5 \cdot \frac{1}{\theta} + 5 \cdot \frac{1}{1-\theta} \cdot (-1) \\ &= \frac{5}{\theta} - \frac{5}{1-\theta}.\end{aligned}$$

Set the derivative equal to zero and solve for θ :

$$\begin{aligned}0 &= \frac{5}{\theta} - \frac{5}{1-\theta} \\ \frac{5}{1-\theta} &= \frac{5}{\theta} \\ \theta &= 1-\theta \\ \theta &= \frac{1}{2}.\end{aligned}$$

To verify that this gives a maximum, rewrite the derivative with a common denominator:

$$\begin{aligned}\frac{dL}{d\theta} &= \frac{5}{\theta} - \frac{5}{1-\theta} \\ &= \frac{5(1-\theta) - 5\theta}{\theta(1-\theta)} \\ &= \frac{5(1-2\theta)}{\theta(1-\theta)}.\end{aligned}$$

For $0 < \theta < 1$, the denominator is positive, so the sign of the derivative is determined by $1 - 2\theta$. Thus the derivative is positive for $0 < \theta < \frac{1}{2}$, zero at $\theta = \frac{1}{2}$, and negative for $\frac{1}{2} < \theta < 1$. The log likelihood therefore increases up to $\theta = \frac{1}{2}$ and decreases afterward. Since the logarithm is strictly increasing, this also maximizes the likelihood on $(0, 1)$. At the endpoints $\theta = 0$ and $\theta = 1$, the likelihood is zero, so neither endpoint gives a larger value.

Therefore the maximum likelihood estimate of θ is $\hat{\theta} = \boxed{\frac{1}{2}}$.

2. (*Question 2*). Let $X_1, X_2, \dots, X_N$ be i.i.d. random variables with density function

$$f(x \mid \sigma) = \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}}.$$

Again, since the observations are independent, we define the likelihood function as

$$\begin{aligned}
l(\sigma \mid x) &\equiv p(x \mid \sigma) = \prod_{t=1}^N p(x_t \mid \sigma) \\
&= \prod_{t=1}^N f(x_t \mid \sigma) \\
&= \prod_{t=1}^N \left(\frac{1}{2\sigma} e^{-\frac{|x_t|}{\sigma}} \right) \\
&= \left(\frac{1}{2\sigma} \right)^N \prod_{t=1}^N \left(e^{-\frac{|x_t|}{\sigma}} \right) \\
&= \left(\frac{1}{2\sigma} \right)^N e^{\left(\sum_{t=1}^N -\frac{|x_t|}{\sigma} \right)} \\
&= \left(\frac{1}{2\sigma} \right)^N e^{\left(-\frac{1}{\sigma} \sum_{t=1}^N |x_t| \right)}.
\end{aligned}$$

We then take the log, which we will optimize.

$$\begin{aligned}
L(\sigma \mid x) &= \log(l(\sigma \mid x)) \\
&= \log \left[\left(\frac{1}{2\sigma} \right)^N e^{-\frac{1}{\sigma} \sum_{t=1}^N |x_t|} \right].
\end{aligned}$$

Applying rules of logarithms we have

$$\begin{aligned}
L(\sigma \mid x) &= \log \left[\left(\frac{1}{2\sigma} \right)^N \right] + \log \left[e^{-\frac{1}{\sigma} \sum_{t=1}^N |x_t|} \right] \\
&= N \log \left(\frac{1}{2\sigma} \right) - \frac{1}{\sigma} \sum_{t=1}^N |x_t| \\
&= -N \log(2\sigma) - \frac{1}{\sigma} \sum_{t=1}^N |x_t| \\
&= -N \log 2 - N \log \sigma - \frac{1}{\sigma} \sum_{t=1}^N |x_t|.
\end{aligned}$$

Taking the derivative of L with respect to σ

$$\begin{aligned}
\frac{dL}{d\sigma} &= 0 - N \cdot \frac{1}{\sigma} - \left(-\frac{1}{\sigma^2}\right) \sum_{t=1}^N |x_t| \\
&= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t|.
\end{aligned}$$

Set the derivative equal to zero and solve for σ :

$$\begin{aligned}
0 &= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t| \\
\frac{N}{\sigma} &= \frac{1}{\sigma^2} \sum_{t=1}^N |x_t|
\end{aligned}$$

Since $\sigma > 0$, we have $\sigma^2 > 0$, so we can multiply both sides by σ^2 :

$$\begin{aligned}
N\sigma &= \sum_{t=1}^N |x_t| \\
\sigma &= \frac{1}{N} \sum_{t=1}^N |x_t|.
\end{aligned}$$

To verify that this gives a maximum, let $\hat{\sigma} = \frac{1}{N} \sum_{t=1}^N |x_t|$ and rewrite the derivative with a common denominator:

$$\begin{aligned}
\frac{dL}{d\sigma} &= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t| \\
&= \frac{\sum_{t=1}^N |x_t| - N\sigma}{\sigma^2} \\
&= \frac{N\hat{\sigma} - N\sigma}{\sigma^2} \\
&= \frac{N(\hat{\sigma} - \sigma)}{\sigma^2}.
\end{aligned}$$

For $\sigma > 0$, the factor N/σ^2 is positive, so the sign of the derivative is determined by $\hat{\sigma} - \sigma$. When $\sum_{t=1}^N |x_t| > 0$, we have $\hat{\sigma} > 0$. Thus the derivative is positive for $0 < \sigma < \hat{\sigma}$, zero at $\sigma = \hat{\sigma}$, and negative for $\sigma > \hat{\sigma}$. The log likelihood therefore increases up to $\hat{\sigma}$ and decreases afterward, so this gives the maximum. Since the logarithm is strictly increasing, maximizing the likelihood is equivalent to maximizing its log. Therefore the maximum likelihood estimate of σ is

$$\begin{aligned}
\hat{\sigma} &= \arg \max_{\sigma > 0} l(x \mid \sigma) \\
&= \arg \max_{\sigma > 0} \log(l(x \mid \sigma)) \\
&= \arg \max_{\sigma > 0} L(\sigma \mid x) \\
&= \boxed{\frac{1}{N} \sum_{t=1}^N |x_t|}.
\end{aligned}$$

3. (*Question 3*). We are given N random samples $X_1, X_2, \dots, X_N$ where we assume that the density function is the Gaussian:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

We use MLE to estimate the parameters μ and σ . I prefer e notation, but the exponents are getting very large for this problem so I am going to switch to $\exp$:

$$\begin{aligned}
l(\mu, \sigma \mid x) &\equiv p(x \mid \mu, \sigma) = \prod_{t=1}^N p(x_t \mid \mu, \sigma) \\
&= \prod_{t=1}^N f(x_t \mid \mu, \sigma^2) \\
&= \prod_{t=1}^N \left[\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_t - \mu)^2}{2\sigma^2}\right) \right] \\
&= \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^N \prod_{t=1}^N \exp\left(-\frac{(x_t - \mu)^2}{2\sigma^2}\right) \\
&= \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^N \exp\left[\sum_{t=1}^N \left(-\frac{(x_t - \mu)^2}{2\sigma^2}\right)\right] \\
&= \frac{1}{(2\pi)^{N/2}\sigma^N} \exp\left[-\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2\right].
\end{aligned}$$

This is old hat at this point, we take the log to make differentiation easier.

$$\begin{aligned}
L(\mu, \sigma \mid x) &= \log(l(\mu, \sigma \mid x)) \\
&= \log \left[\frac{1}{(2\pi)^{N/2} \sigma^N} \exp \left(-\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \right) \right] \\
&= \log \left(\frac{1}{(2\pi)^{N/2} \sigma^N} \right) + \log \left[\exp \left(-\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \right) \right] \\
&= -\log((2\pi)^{N/2} \sigma^N) - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\log((2\pi)^{N/2}) - \log(\sigma^N) - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\frac{N}{2} \log(2\pi) - N \log \sigma - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2.
\end{aligned}$$

Since we are estimating for both μ and σ and L is a function of both, we will have to take partial derivatives. First with respect to μ :

$$\begin{aligned}
\frac{\partial L}{\partial \mu} &= 0 - 0 - \frac{1}{2\sigma^2} \sum_{t=1}^N 2(x_t - \mu) \cdot (-1) \\
&= \frac{1}{\sigma^2} \sum_{t=1}^N (x_t - \mu) \\
&= \frac{1}{\sigma^2} \left(\sum_{t=1}^N x_t - N\mu \right).
\end{aligned}$$

Then with respect to σ :

$$\begin{aligned}
\frac{\partial L}{\partial \sigma} &= 0 - N \cdot \frac{1}{\sigma} - \frac{1}{2} \cdot (-2\sigma^{-3}) \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \mu)^2.
\end{aligned}$$

Now we set these partial derivatives equal to 0 and solve.

$$0 = \frac{1}{\sigma^2} \left(\sum_{t=1}^N x_t - N\mu \right)$$

Since $\sigma > 0$, we can multiply both sides by σ^2 :

$$\begin{aligned} 0 &= \sum_{t=1}^N x_t - N\mu \\ N\mu &= \sum_{t=1}^N x_t \\ \mu &= \boxed{\frac{1}{N} \sum_{t=1}^N x_t = \bar{x}}. \end{aligned}$$

For σ , substituting $\mu = \bar{x}$ gives

$$\begin{aligned} 0 &= -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \bar{x})^2 \\ \frac{N}{\sigma} &= \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \bar{x})^2 \end{aligned}$$

Since $\sigma > 0$, we can multiply both sides by σ^3 :

$$\begin{aligned} N\sigma^2 &= \sum_{t=1}^N (x_t - \bar{x})^2 \\ \sigma^2 &= \frac{1}{N} \sum_{t=1}^N (x_t - \bar{x})^2 \end{aligned}$$

Taking the positive square root, provided the samples are not all identical:

$$\sigma = \boxed{\sqrt{\frac{1}{N} \sum_{t=1}^N (x_t - \bar{x})^2}}.$$

These are, of course, the familiar formulas when working with the normal distribution from statistics.

4. (*Question 4*). Let $X_1, X_2, \dots, X_N$ be an i.i.d. sample from a Poisson distribution with parameter λ and probability mass function

$$P(X = x \mid \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

We use MLE to estimate λ . Again, since the observations are independent, the likelihood is

$$\begin{aligned}
l(\lambda \mid x) &= \prod_{t=1}^N p(x_t \mid \lambda) \\
&= \prod_{t=1}^N \frac{\lambda^{x_t} e^{-\lambda}}{x_t!} \\
&= \frac{\left(\prod_{t=1}^N \lambda^{x_t} \right) \left(\prod_{t=1}^N e^{-\lambda} \right)}{\prod_{t=1}^N x_t!} \\
&= \frac{\lambda^{\sum_{t=1}^N x_t} e^{-N\lambda}}{\prod_{t=1}^N x_t!}.
\end{aligned}$$

We then take the log to make differentiation easier. For $\lambda > 0$, applying rules of logarithms gives

$$\begin{aligned}
L(\lambda \mid x) &= \log(l(\lambda \mid x)) \\
&= \log\left(\frac{\lambda^{\sum_{t=1}^N x_t} e^{-N\lambda}}{\prod_{t=1}^N x_t!}\right) \\
&= \log\left(\lambda^{\sum_{t=1}^N x_t}\right) + \log(e^{-N\lambda}) - \log\left(\prod_{t=1}^N x_t!\right) \\
&= \left(\sum_{t=1}^N x_t\right) \log \lambda - N\lambda - \sum_{t=1}^N \log(x_t!).
\end{aligned}$$

Taking the derivative of L with respect to λ

$$\begin{aligned}
\frac{dL}{d\lambda} &= \left(\sum_{t=1}^N x_t\right) \cdot \frac{1}{\lambda} - N - 0 \\
&= \frac{1}{\lambda} \sum_{t=1}^N x_t - N.
\end{aligned}$$

Set the derivative equal to zero and solve for λ :

$$\begin{aligned}
0 &= \frac{1}{\lambda} \sum_{t=1}^N x_t - N \\
N &= \frac{1}{\lambda} \sum_{t=1}^N x_t
\end{aligned}$$

Since $\lambda > 0$, we can multiply both sides by λ :

$$N\lambda = \sum_{t=1}^N x_t$$

$$\lambda = \frac{1}{N} \sum_{t=1}^N x_t = \bar{x}.$$

To verify that this gives a maximum when $\bar{x} > 0$, rewrite the derivative as

$$\begin{aligned} \frac{dL}{d\lambda} &= \frac{\sum_{t=1}^N x_t - N\lambda}{\lambda} \\ &= \frac{N(\bar{x} - \lambda)}{\lambda}. \end{aligned}$$

For $\lambda > 0$, the factor N/λ is positive, so the sign of the derivative is determined by $\bar{x} - \lambda$. Thus the derivative is positive for $0 < \lambda < \bar{x}$, zero at $\lambda = \bar{x}$, and negative for $\lambda > \bar{x}$. The log likelihood therefore increases up to $\bar{x}$ and decreases afterward. Since the logarithm is strictly increasing, this also maximizes the likelihood.

Therefore, on the parameter space $\lambda > 0$, the maximum likelihood estimate is

$$\begin{aligned} \hat{\lambda} &= \arg \max_{\lambda > 0} l(\lambda \mid x) \\ &= \arg \max_{\lambda > 0} L(\lambda \mid x) \\ &= \boxed{\frac{1}{N} \sum_{t=1}^N x_t = \bar{x}}. \end{aligned}$$

5. (*Question 5*). We are given N data samples $(X_1, Y_1), (X_2, Y_2), \dots, (X_N, Y_N)$. For each input X_i , the neural network predicts both a mean $\mu(X_i)$ and a standard deviation $\sigma(X_i)$. We assume that the output Y follows a Gaussian distribution conditioned on X :

$$Y \mid X \sim \mathcal{N}(\mu(X), \sigma^2(X)).$$

Therefore, for a single sample (X_i, Y_i) , the conditional density is

$$p(Y_i \mid X_i) = \frac{1}{\sqrt{2\pi}\sigma(X_i)} \exp\left(-\frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)}\right).$$

As before, assuming the samples are independent, the likelihood of observing all N samples is the product of the individual probabilities:

$$\begin{aligned}
l &= \prod_{i=1}^N p(Y_i | X_i) \\
&= \prod_{i=1}^N \left[\frac{1}{\sqrt{2\pi}\sigma(X_i)} \exp \left(-\frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right) \right].
\end{aligned}$$

We could maximize this likelihood directly, but as before it is easier to work with the log likelihood:

$$\begin{aligned}
L &= \log l \\
&= \sum_{i=1}^N \log \left[\frac{1}{\sqrt{2\pi}\sigma(X_i)} \exp \left(-\frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right) \right] \\
&= \sum_{i=1}^N \left[\log \left(\frac{1}{\sqrt{2\pi}\sigma(X_i)} \right) - \frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right] \\
&= \sum_{i=1}^N \left[-\frac{1}{2} \log(2\pi) - \log \sigma(X_i) - \frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right].
\end{aligned}$$

Maximum Likelihood Estimation seeks to maximize this quantity. However, the loss function for the neural network is to be minimized, so we instead minimize the negative log likelihood:

$$\mathcal{L} = -L.$$

Therefore,

$$\mathcal{L} = \sum_{i=1}^N \left[\frac{1}{2} \log(2\pi) + \log \sigma(X_i) + \frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right].$$

Equivalently, using $\log(\sigma^2) = 2 \log \sigma$,

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^N \left[\log(2\pi\sigma^2(X_i)) + \frac{(Y_i - \mu(X_i))^2}{\sigma^2(X_i)} \right].$$

6. (Question 6). Code for this problem can be found in `q6_cascade_forest.ipynb`.

A deep forest is used for a classification problem in which each observation is represented by a vector of d features and belongs to one of k possible classes.

Initially, we feed the vector of d features into the first layer of the deep forest. In the architecture shown, each layer contains two random forests and two completely random forests.

A random forest contains many decision trees that are made different from one another by introducing randomness in the training samples and in the features considered when choosing splits. In a completely random forest, a random feature and a random threshold are selected at each node rather than searching for the best split.

Each tree produces probabilities for the k possible classes. The probability vectors produced by the trees in a forest are aggregated into a single vector of length k containing that forest’s estimated probabilities for each possible class.

Since the layer contains four forests, it therefore produces four vectors of length k . These four class-probability vectors are concatenated with the original d features, producing a new feature vector of length $d + 4k$.

This augmented feature vector is then passed to the next layer, which repeats the same process. Each layer therefore receives both the original features and the class-probability information produced by the immediately preceding layer.

We keep the probability vectors and not just the predicted class because the vectors contain useful information about how likely the other possible classes are. For example, a forest might predict that an observation is walking while also indicating that walking upstairs is plausible but sitting down is extremely unlikely. This information would be lost if only the most likely class were passed to the next layer.

The cascade continues through additional layers, with the class-probability vectors becoming increasingly informed by the outputs of the previous layers. At the final layer, the probability vectors produced by the forests are combined, and the class with the largest resulting probability is taken as the final prediction.

For this particular experiment, we used the UCI Human Activity Recognition dataset, which contains 10,299 observations represented by 561 numeric features and divided among six activity classes. We combined the original training and test data and created a new 80:10:10 split, shown in Table 1. The validation set was used to choose the depth of the cascade, while the test set was reserved for the final comparison.

Set	Observations	Fraction
Training	8,239	80%
Validation	1,030	10%
Test	1,030	10%
Total	10,299	100%

Table 1: Training, validation, and test split for the UCI HAR dataset.

Our implementation used four forests at each cascade layer: two standard random forests and two randomized forests approximating the completely random forests in the given architecture. Each forest contained 100 trees. Since there are six possible classes, each forest produced a vector of six class probabilities. The four probability vectors gave 24 additional features, so after the first layer the input to each subsequent layer consisted of the original 561 features together with the 24 probability features from the previous layer,

for a total of 585 features. During training, three-fold out-of-fold predictions were used to generate the probability features passed to the next layer.

We allowed the cascade to grow to a maximum of four layers and selected the depth using validation accuracy. Validation accuracy increased from 0.9786 at one layer to 0.9864 at two layers and 0.9884 at three layers. The fourth layer did not improve validation accuracy, so the selected cascade depth was three.

The final results are shown in Table 2. The standard random forest achieved a test accuracy of 0.9806, while the three-layer cascade forest achieved a test accuracy of 0.9893. Thus, in this experiment the cascade improved test accuracy by approximately 0.874 percentage points.

Model	Validation Acc.	Test Acc.	Macro F1	Trees
Standard Random Forest	0.9825	0.9806	0.9798	400
Cascade Forest	0.9884	0.9893	0.9895	1,200

Table 2: Performance of the standard random forest and the selected three-layer cascade forest.

The improvement came at additional computational cost. The standard random forest trained in approximately 7.3 seconds, while the cascade depth-selection process took approximately 47.3 seconds. The gains in accuracy are probably not worth the nearly 7x increase in training time.

7. (*Question 7*). In this problem we use the information gain method to build a decision tree for the table below.

x_1	x_2	y
0	1	0
1	1	0
0	0	1
1	0	0
0	1	0
0	1	1

We begin by calculating the entropy of the class label Y . There are four examples with $y = 0$ and two examples with $y = 1$, so

$$P(Y = 0) = \frac{4}{6}, \quad P(Y = 1) = \frac{2}{6}.$$

Therefore,

$$\begin{aligned} H(Y) &= -\frac{4}{6} \log_2 \left(\frac{4}{6} \right) - \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \\ &\approx 0.9183. \end{aligned}$$

We next calculate the conditional entropy after splitting on x_1 .

For $x_1 = 0$, there are four examples with class labels

$$0, 1, 0, 1,$$

so

$$P(Y = 0 \mid x_1 = 0) = \frac{2}{4}, \quad P(Y = 1 \mid x_1 = 0) = \frac{2}{4},$$

and

$$\begin{aligned} H(Y \mid x_1 = 0) &= -\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \\ &= 1. \end{aligned}$$

For $x_1 = 1$, there are two examples and both have $y = 0$. Thus,

$$H(Y \mid x_1 = 1) = 0.$$

The conditional entropy for x_1 is therefore

$$\begin{aligned} H(Y \mid X_1) &= P(x_1 = 0)H(Y \mid x_1 = 0) + P(x_1 = 1)H(Y \mid x_1 = 1) \\ &= \frac{4}{6}(1) + \frac{2}{6}(0) \\ &= \frac{2}{3} \approx 0.6667. \end{aligned}$$

Thus, the information gain from splitting on x_1 is

$$\begin{aligned} IG(X_1) &= H(Y) - H(Y \mid X_1) \\ &= 0.9183 - 0.6667 \\ &\approx 0.2516. \end{aligned}$$

We now perform the same calculation for x_2 .

For $x_2 = 0$, there are two examples, one with $y = 0$ and one with $y = 1$. Therefore,

$$P(Y = 0 \mid x_2 = 0) = \frac{1}{2}, \quad P(Y = 1 \mid x_2 = 0) = \frac{1}{2},$$

and

$$H(Y \mid x_2 = 0) = -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) = 1.$$

For $x_2 = 1$, there are four examples, three with $y = 0$ and one with $y = 1$. Thus,

$$P(Y = 0 \mid x_2 = 1) = \frac{3}{4}, \quad P(Y = 1 \mid x_2 = 1) = \frac{1}{4},$$

and

$$\begin{aligned} H(Y \mid x_2 = 1) &= -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \\ &\approx 0.8113. \end{aligned}$$

Therefore,

$$\begin{aligned} H(Y \mid X_2) &= P(x_2 = 0)H(Y \mid x_2 = 0) + P(x_2 = 1)H(Y \mid x_2 = 1) \\ &= \frac{2}{6}(1) + \frac{4}{6}(0.8113) \\ &\approx 0.8742. \end{aligned}$$

The information gain from splitting on x_2 is therefore

$$\begin{aligned} IG(X_2) &= H(Y) - H(Y \mid X_2) \\ &= 0.9183 - 0.8742 \\ &\approx 0.0441. \end{aligned}$$

Since

$$IG(X_1) > IG(X_2),$$

x_1 is selected as the root node.

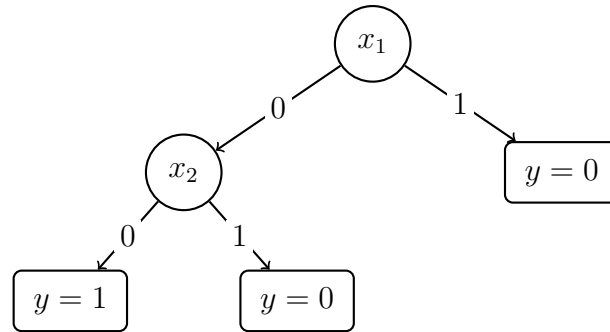
For the branch $x_1 = 1$, both examples have $y = 0$, so this branch is already pure and becomes a leaf with class $y = 0$.

For the branch $x_1 = 0$, the class labels are not pure, so we split again using the remaining attribute x_2 . If $x_2 = 0$, the only example has $y = 1$, so this becomes a leaf with class $y = 1$. If $x_2 = 1$, the three class labels are

$$0, 0, 1.$$

No attributes remain for further splitting, so majority voting is used and this leaf is assigned class $y = 0$.

The resulting decision tree is



The final leaf for $x_1 = 0$ and $x_2 = 1$ is not pure because the identical input $(x_1, x_2) = (0, 1)$ occurs with both class labels in the training data. Since no additional attributes are available, the majority class $y = 0$ is used.

8. (*Question 8*). Solutions and code for this question can be found in `q8_classification_metrics.ipynb`.